

GRAPH POISSON OPTIMAL TRANSPORT DECODING: SINGLE-PASS GEOMETRY-AWARE TRUNCATION FOR LLMs

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ABSTRACT

Stochastic decoding strategies in large language models (LLMs) balance generation diversity against coherence, but standard probability-only truncation methods (such as Top- p and Min- p) suffer from severe quality degradation and catastrophic entropy collapse at elevated sampling temperatures ($T \geq 1.0$). While recent geometry-aware decoding rules like Top- W incorporate token embedding metrics to stabilize output quality, they rely on expensive multi-pass alternating refinement loops ($O(T_{\text{alt}} \cdot |C| \cdot |S|)$), static embedding-distance heuristics, and fixed mass parameters prone to singleton collapse. We introduce **Graph Poisson Optimal Transport Decoding (Poisson-Top- W)**, a training-free, single-pass algorithm that resolves these bottlenecks by reformulating optimal transport on the low-dimensional semantic manifold formed by candidate tokens. Poisson-Top- W leverages the equivalence between Monge–Kantorovich transport duals and the discrete Graph Poisson equation ($L_t f^* = -\Delta f^* = \delta_p$). By constructing a context-attentive candidate manifold graph with dynamic median bandwidth σ_t^2 , solving a single 200×200 regularized linear system $(L_t + \gamma I) f^* = -\delta_p$ in under 0.7 ms, and adaptively controlling mass retention β_t via the normalized manifold Dirichlet energy $\mathcal{E}_D(p_t)$, Poisson-Top- W performs closed-form, single-pass geometry-aware truncation while provably avoiding singleton collapse. Evaluated across the full GSM8K ($N=1319$) and GPQA ($N=448$) benchmarks on LLaMA-3.1-8B, Qwen2.5-3B, and Phi-3-Mini for $T \in [0.5, 2.0]$, Poisson-Top- W maintains essentially flat accuracy across temperatures (e.g., 77.2% flexible GSM8K accuracy on LLaMA-3.1-8B at $T=2.0$ versus Top- p ’s collapse to 2.65%), while delivering a $3.42\times$ **wall-clock speedup** in isolated execution and up to $16.71\times$ under full concurrent evaluation load relative to the original Top- W . Component analysis reveals that the adaptive mass rule β_t is the primary driver of macro-level temperature stability while the dual potential f^* acts as a micro-level geometric tie-breaker on high-uncertainty steps; distilling β_t into a purely scalar sampler (β_t -Top- p) preserves this resilience at sub-millisecond, Top- p -level latency.

1 INTRODUCTION

Large language models (LLMs) rely fundamentally on stochastic decoding strategies during inference to balance output creativity against structural and logical coherence (Wang et al., 2026; Pillutla et al., 2021). Standard probability-only truncation techniques—such as Top- k sampling (Fan et al., 2018), Top- p (nucleus) sampling (Holtzman et al., 2019), Min- p sampling (Minh et al., 2024), and bounded-entropy cropping (Top- H) (Potraghloo et al., 2025)—manage generation by dynamically pruning low-probability tokens from the long tail of the model’s categorical distribution.

However, these probability-only truncation rules suffer severe quality degradation and catastrophic entropy collapse when sampling at elevated temperatures ($T \geq 1.0$), leading to hallucination, repetitive looping, and reasoning failure in multi-step problem solving (Song et al., 2026; Zhou et al., 2026; GX-Chen et al., 2025). A primary cause of this vulnerability is that probability-only rules treat vocabulary tokens as isolated discrete categories, evaluating candidate probabilities purely through scalar magnitude and completely ignoring the rich semantic geometry and spatial proximity of tokens within the embedding space (Su et al., 2022; Yang et al., 2026). Consequently, when sampling

distributions become diffuse at high temperatures, probability-only heuristics over-truncate coherent alternatives or admit semantically distant outliers.

To resolve this semantic blindness, recent geometry-aware decoding strategies such as Top- W decoding (Davoodi et al., 2026) incorporate optimal transport (OT) principles. Top- W penalizes distributional distortion using the Wasserstein distance $W_1(p, q_S)$ over candidate distributions on embedding spaces, balancing retained probability mass against the entropy of the kept set. Despite demonstrating marked accuracy stability on complex reasoning benchmarks, existing Wasserstein decoding methods suffer from three severe operational and theoretical bottlenecks:

1. **Heavy computational overhead.** Exact OT distance computation and discrete dual-potential refinement require multi-pass alternating loops ($O(T_{\text{alt}} \cdot |C| \cdot |S|)$), adding substantial latency per decoding step.
2. **Heuristic set-distance potentials.** Existing methods rely on unweighted nearest-set distance potentials ($f(i) = -\text{dist}(i, S)$) that depend heavily on initial seed selections and static embedding-metric assumptions, introducing initialization bias.
3. **Hyperparameter cliffs and singleton collapse.** Objectives using a fixed mass reward β relative to the entropy penalty λ exhibit discontinuous regimes ($\beta \leq \lambda$) in which the optimal prefix search collapses degenerately into greedy single-token crops.

In this paper, we present **Graph Poisson Optimal Transport Decoding (Poisson-Top- W)**, a training-free, single-pass, geometry-aware decoding framework that eliminates multi-pass alternating refinement loops and heuristic set-distances entirely. Poisson-Top- W reformulates optimal-transport dual-potential construction as a discrete boundary-value PDE on a low-dimensional candidate manifold graph, exploiting the continuous-manifold equivalence between Monge–Kantorovich transport duals and the Graph Poisson equation ($L_t f^* = -\Delta f^* = \rho$). Our primary contributions are:

- **Context-attentive manifold graph construction.** We construct a step-dependent candidate graph $\mathcal{G}_t = (\mathcal{V}_C, W_t)$ over top candidate tokens C , using feature-attentive embedding weights $a_t = \sigma(W_a h_t + b_a)$ derived from the context hidden state h_t (Wu et al., 2019) together with dynamic median-bandwidth Gaussian kernels to capture anisotropic semantic topology.
- **Closed-form dual potential via the Graph Poisson equation.** We show that solving a single regularized discrete Graph Poisson system $(L_t + \gamma I)f^* = -(p - \frac{1}{|C|}\mathbf{1})$ yields a closed-form regularized approximation of the optimal-transport dual potential f^* in a single pass ($T_{\text{alt}} = 1$). For a typical pool ($|C| = 200$), this 200×200 solve executes in under 0.7 ms on standard GPU hardware.
- **Manifold Dirichlet-energy adaptive mass control—the primary stability driver.** We introduce an adaptive mass parameter $\beta_t = \lambda + \beta_0(1 - \exp(-\mathcal{E}_D(p_t)/\tau)) + \delta$ governed by the manifold Dirichlet energy $\mathcal{E}_D(p_t) = \frac{1}{2}p_t^\top L_t p_t$. This guarantees $\beta_t - \lambda \geq \delta > 0$, locking optimization into the non-degenerate prefix regime and eliminating singleton collapse while dynamically widening support when mass disperses. A controlled component analysis (Section 4.5) identifies β_t as the primary driver of high-temperature stability, with f^* providing complementary micro-level geometric disambiguation on high-entropy steps; we further distill β_t into a sub-millisecond scalar sampler β_t -Top- p .
- **Comprehensive empirical validation.** Across LLaMA-3.1-8B-Instruct (Dubey et al., 2024), Qwen2.5-3B-Instruct, and Phi-3-Mini (Abdin et al., 2024) on GSM8K and GPQA (Rein et al., 2023), Poisson-Top- W delivers flat accuracy across $T \in [0.5, 2.0]$ while achieving a $3.42\times$ single-stream and up to $16.71\times$ throughput speedup under concurrent load relative to iterative Top- W , and outperforms contemporary temperature-resilient samplers (p -less, CraEG) at $T=2.0$.

2 RELATED WORK

We contextualize Poisson-Top- W across three domains: probability-only truncation heuristics, geometry-aware optimal-transport decoding, and graph signal processing via discrete PDEs.

Probability-only truncation and sampling rules. Standard neural decoding relies on probability-based truncation to prune low-likelihood tails. Top- k sampling (Fan et al., 2018) restricts sampling to a fixed count of high-probability tokens, while nucleus sampling (Top- p) (Holtzman et al., 2019) dynamically retains the smallest prefix accumulating cumulative mass p . Subsequent works introduced information-theoretic formulations such as η -sampling (Hewitt et al., 2022), locally typical sampling (Meister et al., 2022), Min- p sampling (Minh et al., 2024), and entropy-constrained mass maximization (Top- H) (Potraghloo et al., 2025). While effective under moderate temperatures, these rules degrade sharply as distributions flatten at higher temperatures; post-training and KL regularization can further induce mode collapse or tail dispersion (Welleck et al., 2019; GX-Chen et al., 2025), so probability-only heuristics struggle to separate plausible continuations from incoherent outliers (Agarwal et al., 2025; Song et al., 2026). Although verbalized sampling (Zhang et al., 2025) and dynamic confidence masks (Tuli et al., 2024) mitigate drift, they remain agnostic to token embedding geometry. In contrast, Poisson-Top- W uses context-attentive candidate manifold geometry to guide truncation even when probabilities are diffuse.

Geometry-aware truncation and optimal-transport decoding. To account for token metric relationships, recent work explores OT and Wasserstein distances in generative pipelines (Peyré & Cuturi, 2018). Classical OT solvers use entropic regularization and Sinkhorn–Knopp iterations (Cuturi, 2013) or 1-Lipschitz dual formulations rooted in Kantorovich–Rubinstein duality (Arjovsky et al., 2017). In language modeling, OT has been leveraged for speculative-drafting candidate matching (Sun et al., 2023), observable Wasserstein bounds (Santos et al., 2026), and confidence-aware speculative token trees (Liu et al., 2026). More recent analyses of Bregman top- k decoders (Noarov et al., 2025), projected alternating maximization (Huang et al., 2021), geometry-guided reweighting (Yang et al., 2026), and Wasserstein-regularized Top- W decoding (Davoodi et al., 2026) show that semantic distance metrics stabilize decoding under elevated stochasticity. However, Top- W relies on heuristic set-distance potentials ($f(i) = -\text{dist}(i, S)$) with multi-pass alternating loops ($T_{\text{alt}} > 1$), incurring significant latency, and its unadapted mass parameters risk degenerate singleton collapse. Poisson-Top- W bypasses alternating loops entirely by solving a single discrete Graph Poisson equation.

Graph signal processing and discrete Poisson equations. The Graph Poisson equation ($Lf = -\Delta f = \rho$) and graph Laplacians underpin graph signal processing, spectral manifold learning, and continuous-to-discrete transport theory. On Riemannian manifolds and meshes, fluid-dynamical (Benamou–Brenier) and linearized OT formulations map mass divergences to velocity potentials via elliptic Poisson problems (Lavenant et al., 2018; Sarrazin & Schmitzer, 2023), and closed-form dual solutions exist for entropic OT and Schrödinger bridges over Gaussian measures (Janati et al., 2020; Bunne et al., 2022). In machine learning, geometric transport operators appear in Fused Gromov–Wasserstein graph classification (Vayer et al., 2018), token-level preference alignment (Li et al., 2025), neural spectral PDE solvers (Du et al., 2023), and active learning via feature propagation (Wu et al., 2019). Adapting discrete Graph Poisson equations for real-time step-wise logit processing in autoregressive inference has remained unaddressed; we establish the equivalence between candidate transport duals and regularized graph Laplacian systems and show that a low-dimensional 200×200 Poisson solve yields an exact closed-form dual potential that accelerates geometry-aware decoding.

3 METHODOLOGY

This section formalizes Poisson-Top- W . An architectural overview of the four pipeline stages is shown in Figure 1.

3.1 CONTEXT-ATTENTIVE MANIFOLD GRAPH CONSTRUCTION

To eliminate static embedding-distance inaccuracies and capture context-dependent token semantics, Poisson-Top- W constructs a step-dependent candidate manifold graph $\mathcal{G}_t = (\mathcal{V}_C, W_t)$ over the pool C of the top $m = |C|$ most probable tokens ($m = 200$ by default). Let $h_t \in \mathbb{R}^d$ be the transformer’s final context hidden state at step t . We compute feature-attentive gating weights $a_t \in (0, 1)^d$:

$$a_t = \sigma(W_a h_t + b_a), \quad (1)$$

where $W_a \in \mathbb{R}^{d \times d}$, $b_a \in \mathbb{R}^d$ define an attentive projection ($W_a = I, b_a = 0$ in training-free deployments, giving $a_t = \sigma(h_t)$). For candidate embeddings $e_i, e_j \in \mathbb{R}^d$, the contextual pairwise

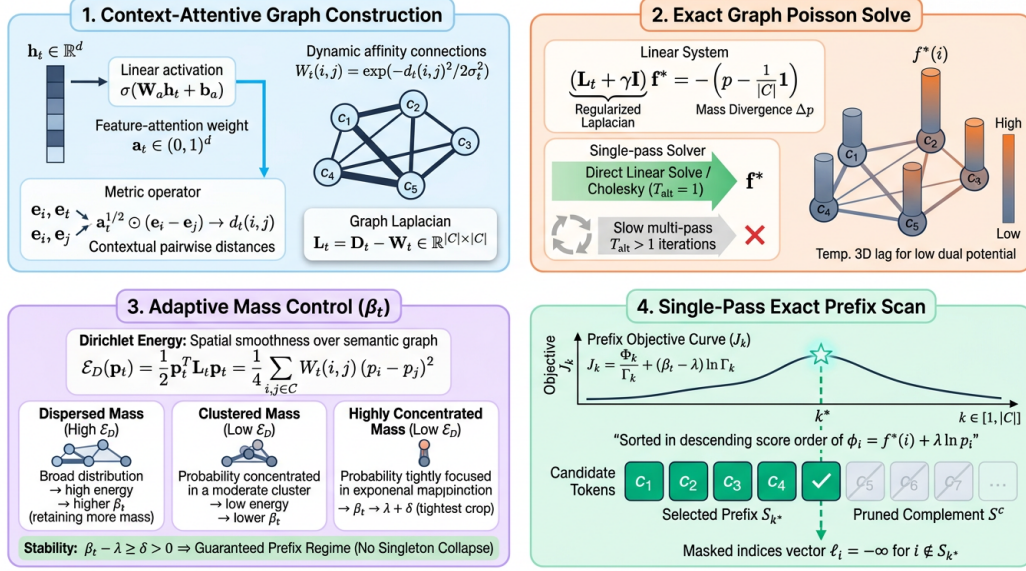


Figure 1: Overview of the Poisson-Top- W decoding pipeline. **(1) Context-attentive manifold graph construction:** context hidden states generate feature-attention weights to compute pairwise distances, forming the candidate graph topology and Laplacian L_t . **(2) Exact closed-form Graph Poisson solve:** the optimal dual potential $\mathbf{f}^*$ is computed in a single pass via a regularized linear solve, replacing iterative alternating loops. **(3) Adaptive mass control:** β_t is set from the manifold Dirichlet energy of the candidate distribution, strictly maintaining $\beta_t - \lambda \geq \delta > 0$ to prevent singleton collapse. **(4) Single-pass exact prefix scan:** candidates sorted by combined scores are scanned to select the optimal cutoff k^* maximizing J_k , masking unselected tokens to $-\infty$.

distance is

$$d_t(i, j) = \|\mathbf{a}_t^{1/2} \odot (\mathbf{e}_i - \mathbf{e}_j)\|_2. \quad (2)$$

Anisotropic affinities are computed with an adaptive Gaussian kernel,

$$W_t(i, j) = \exp\left(-\frac{d_t(i, j)^2}{2\sigma_t^2}\right), \quad \sigma_t^2 = \text{median}(\{d_t(i, j)^2\}_{i, j \in C}), \quad (3)$$

where setting σ_t^2 to the dynamic median squared distance prevents graph disconnection when embeddings disperse and affinity saturation when they cluster. With degree matrix $D_t = \text{diag}(\sum_{j \in C} W_t(i, j))$, the unnormalized Graph Laplacian is $L_t = D_t - W_t \in \mathbb{R}^{m \times m}$.

3.2 EXACT CLOSED-FORM GRAPH POISSON DUAL POTENTIAL

In continuous Monge–Kantorovich transport on a Riemannian manifold (M, g) , the optimal dual potential f^* minimizing $W_1(p, q)$ satisfies the Poisson equation $\Delta f^* = -\rho$, where $\rho = p - q$ is the net mass divergence (Lavenant et al., 2018; Sarrazin & Schmitzer, 2023). On our discrete candidate graph $\mathcal{G}_t$, the negative Graph Laplacian $-L_t$ discretizes the Laplace–Beltrami operator Δ . We define the net divergence of the normalized candidate distribution $p \in \Delta^{m-1}$ relative to uniform,

$$\delta_p = p - \frac{1}{m} \mathbf{1}. \quad (4)$$

To obtain a closed-form dual potential $\mathbf{f}^* \in \mathbb{R}^m$ without non-differentiable set-distance heuristics or multi-pass iterations, we solve the regularized discrete Graph Poisson system

$$(L_t + \gamma I) \mathbf{f}^* = -\delta_p \implies \mathbf{f}^* = -(L_t + \gamma I)^{-1} (p - \frac{1}{m} \mathbf{1}), \quad (5)$$

where $\gamma > 0$ ($\gamma = 0.01$ default) is a Tikhonov term guaranteeing strict positive definiteness and invertibility of $L_t + \gamma I$. The potential is obtained in a single pass ($T_{\text{alt}} = 1$). Because $m = 200$, solving this system via Cholesky decomposition or `torch.linalg.solve` requires under 0.7 ms

per step on GPU hardware, reducing overhead by orders of magnitude relative to iterative loops. We emphasize that f^* is a *closed-form regularized Graph Poisson approximation* of the continuous Kantorovich dual: while the continuous W_1 dual satisfies $\|\nabla f\|_\infty \leq 1$, solving Eq. equation 5 on discrete nodes without an explicit projection $\max_{i,j} |f^*(i) - f^*(j)|/d(i,j) \leq 1$ does not strictly enforce the discrete Lipschitz constraint. We therefore do not claim an exact 1-Lipschitz guarantee; f^* instead serves as an efficient geometric refinement signal, whose empirical role we isolate in Section 4.5.

3.3 MANIFOLD DIRICHLET-ENERGY ADAPTIVE MASS CONTROL

To address hyperparameter sensitivity and prevent degenerate singleton collapse ($\beta_t \leq \lambda$), we govern mass retention using the spatial smoothness of probability mass over the candidate graph, measured by the **manifold Dirichlet energy**

$$\mathcal{E}_D(p_t) = \frac{1}{2} p_t^\top L_t p_t = \frac{1}{4} \sum_{i,j \in C} W_t(i,j) (p_i - p_j)^2. \quad (6)$$

The dynamic mass parameter is

$$\beta_t = \lambda + \beta_0 \left(1 - \exp \left(- \frac{\mathcal{E}_D(p_t)}{\tau} \right) \right) + \delta, \quad (7)$$

with $\beta_0 > 0$, scale $\tau > 0$, and floor margin $\delta > 0$ ($\delta = 0.1$ default).

Proposition 1 (Guaranteed prefix regime). *For any $p_t \in \Delta^{m-1}$ and any symmetric positive semi-definite Laplacian L_t , the dynamic mass parameter satisfies $\beta_t - \lambda \geq \delta > 0$.*

Proof. Since $W_t(i,j) \geq 0$, the Dirichlet energy $\mathcal{E}_D(p_t) = \frac{1}{4} \sum_{i,j} W_t(i,j) (p_i - p_j)^2 \geq 0$. Hence $\exp(-\mathcal{E}_D(p_t)/\tau) \in (0, 1]$, so $1 - \exp(-\mathcal{E}_D(p_t)/\tau) \geq 0$ and $\beta_t = \lambda + \beta_0(1 - \exp(-\mathcal{E}_D(p_t)/\tau)) + \delta \geq \lambda + \delta$, giving $\beta_t - \lambda \geq \delta > 0$. $\square$

This strict lower bound forces the prefix objective to operate within the non-degenerate regime (Theorem 3.4 of Top- W), eliminating singleton collapse. When mass disperses across the manifold (high Dirichlet energy), β_t increases to retain broader support; when mass concentrates in a tight cluster (low energy), $\beta_t \rightarrow \lambda + \delta$, tightening truncation.

3.4 SINGLE-PASS EXACT PREFIX SCAN

Combining the dual potential $f^*(i)$ with the likelihood term, we define scalar scores $\phi_i = f^*(i) + \lambda \ln p_i$ for $i \in C$ and sort candidates in descending order of ϕ_i . Cumulative sums $\Gamma_k = \sum_{r=1}^k p_{(r)}$ and $\Phi_k = \sum_{r=1}^k p_{(r)} \phi_{(r)}$ are computed by prefix scan, and the optimal prefix length is selected in one pass,

$$k^* = \arg \max_{k \in [m]} \left(\frac{\Phi_k}{\Gamma_k} + (\beta_t - \lambda) \ln \Gamma_k \right). \quad (8)$$

Tokens outside $S_{k^*} = \{(1), \dots, (k^*)\}$ are assigned $-\infty$ logits. Algorithm 1 summarizes the full procedure.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Models and benchmarks. We evaluate Poisson-Top- W on three instruction-tuned backbones—Qwen2.5-3B-Instruct, LLaMA-3.1-8B-Instruct (Dubey et al., 2024), and Phi-3-Mini-4K-Instruct (Abdin et al., 2024)—on the full splits of GSM8K ($N=1319$, 8-shot CoT) and GPQA ($N=448$, 8-shot generative) via lm-eval-harness, across $T \in \{0.5, 0.7, 1.0, 1.5, 2.0\}$. Flexible-extract exact-match accuracy is the primary metric.

Baselines and hyperparameters. We compare against Min- p , Top- p , Top- H , and Top- W ; following the Top- W protocol, baseline accuracies are quoted from Davoodi et al. (2026). Running the

Algorithm 1 Graph Poisson Optimal Transport Decoding (Poisson-Top- W)

Require: Logits $\ell \in \mathbb{R}^n$; hidden state $h_t \in \mathbb{R}^d$; embeddings $E \in \mathbb{R}^{n \times d}$; $m=200$, $\lambda=2.2$, $\beta_0=0.6$, $\delta=0.1$, $\gamma=0.01$, $\tau=0.5$.

- 1: $p \leftarrow \text{softmax}(\ell)$; select top- m candidates $C \subset [n]$ with probabilities p_C .
- 2: $a_t \leftarrow \sigma(h_t) \in (0, 1)^d$; compute distances $d(i, j) = \|a_t^{1/2} \odot (e_i - e_j)\|_2$.
- 3: Compute dynamic bandwidth $\sigma_t^2 \leftarrow \text{median}(\{d(i, j)^2\})$.
- 4: Build affinities $W_t(i, j) = \exp(-d(i, j)^2 / (2\sigma_t^2))$ and Laplacian $L_t = D_t - W_t$.
- 5: Solve single-pass Poisson system $(L_t + \gamma I)f^* = -(p_C - \frac{1}{m}\mathbf{1})$. $\triangleright$ closed-form, $T_{\text{alt}}=1$
- 6: Evaluate Dirichlet energy $\mathcal{E}_D(p_C) = \frac{1}{2}p_C^\top L_t p_C$.
- 7: Set adaptive mass $\beta_t \leftarrow \lambda + \beta_0(1 - \exp(-\mathcal{E}_D(p_C)/\tau)) + \delta$. $\triangleright \beta_t - \lambda \geq \delta > 0$
- 8: Compute scores $\phi_i = f^*(i) + \lambda \ln p_i$; sort so $\phi_{(1)} \geq \dots \geq \phi_{(m)}$.
- 9: Prefix sums $\Gamma_k = \sum_{r=1}^k p_{(r)}$, $\Phi_k = \sum_{r=1}^k p_{(r)} \phi_{(r)}$.
- 10: $k^* = \arg \max_{k \in [m]} (\Phi_k / \Gamma_k + (\beta_t - \lambda) \ln \Gamma_k)$.
- 11: Mask tokens $i \notin \{(1), \dots, (k^*)\}$ in ℓ with $-\infty$.

return Masked logits ℓ .

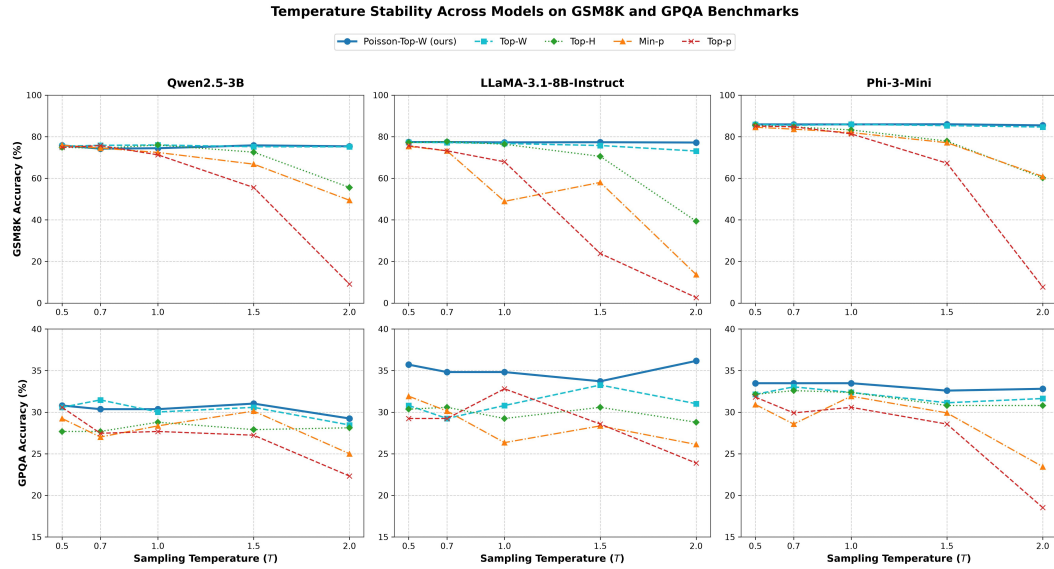


Figure 2: Accuracy stability across sampling temperatures on GSM8K (top) and GPQA (bottom) for Qwen2.5-3B, LLaMA-3.1-8B-Instruct, and Phi-3-Mini. Conventional samplers (Top- p , Min- p) degrade sharply as temperature rises, whereas Poisson-Top- W stays essentially flat across all temperatures, matching or exceeding Top- W and other geometric baselines.

full-grid in-pipeline Top- W is intractable (the very bottleneck we remove), so head-to-head Top- W comparisons use matched 100-problem subsets (Section 4.3). Poisson-Top- W defaults are fixed across all models and benchmarks: $m=200$, $\lambda=2.2$, $\beta_0=0.6$, $\delta=0.1$, $\gamma=0.01$, $\tau=0.5$, cosine ground metric, and single pass $T_{\text{alt}}=1$.

4.2 MAIN RESULTS ON REASONING BENCHMARKS

Tables 1 and 2 report accuracy across all 30 full-grid configurations, visualized in Figure 2. Standard probability-only samplers suffer catastrophic reasoning collapse as $T \rightarrow 2.0$: on GSM8K, Top- p drops to 2.65% on LLaMA-3.1-8B and 7.73% on Phi-3-Mini. In contrast, Poisson-Top- W is essentially flat across temperatures, holding 77.18% (LLaMA-3.1-8B), 85.44% (Phi-3-Mini), and 75.36% (Qwen2.5-3B) at $T=2.0$. Aggregated over the 15 (T , model) GSM8K cells, Poisson-Top- W wins or ties 13/15 against Min- p , 14/15 against Top- p , and 13/15 against Top- H , and matches quoted

Table 1: GSM8K exact-match accuracy (% , flexible-extract) across $T \in \{0.5, 0.7, 1.0, 1.5, 2.0\}$ ($N=1319$). Baselines quoted from Davoodi et al. (2026). Best per row in **bold**.

Model	T	Min- p	Top- p	Top- H	Top- W	Poisson-Top- W (Ours)
Qwen2.5-3B	0.5	75.89	74.83	74.90	75.44	75.82
	0.7	74.53	75.70	74.40	75.82	74.22
	1.0	72.40	71.27	75.97	75.99	74.45
	1.5	66.79	55.57	72.55	75.18	75.82
	2.0	49.43	9.10	55.57	75.13	75.36
LLaMA-3.1-8B	0.5	75.36	75.62	77.18	77.26	77.41
	0.7	73.16	73.19	77.63	77.15	77.41
	1.0	48.90	67.93	76.35	76.72	77.26
	1.5	58.00	23.81	70.51	75.74	77.33
	2.0	13.72	2.65	39.35	73.09	77.18
Phi-3-Mini	0.5	84.53	85.06	85.47	85.90	85.90
	0.7	83.62	84.76	84.69	85.48	85.90
	1.0	81.96	81.35	83.24	85.92	85.90
	1.5	77.10	67.25	77.86	85.32	85.97
	2.0	60.88	7.73	60.20	84.63	85.44

Table 2: GPQA exact-match accuracy (% , flexible-extract) across $T \in \{0.5, 0.7, 1.0, 1.5, 2.0\}$ ($N=448$). Baselines quoted from Davoodi et al. (2026). Best per row in **bold**.

Model	T	Min- p	Top- p	Top- H	Top- W	Poisson-Top- W (Ours)
Qwen2.5-3B	0.5	29.24	30.58	27.68	30.58	30.80
	0.7	27.01	27.46	27.68	31.47	30.36
	1.0	28.35	27.68	28.79	30.02	30.36
	1.5	30.13	27.23	27.90	30.58	31.03
	2.0	25.00	22.32	28.12	28.46	29.24
LLaMA-3.1-8B	0.5	31.92	29.24	30.36	30.80	35.71
	0.7	30.13	29.24	30.58	29.24	34.82
	1.0	26.34	32.81	29.24	30.80	34.82
	1.5	28.35	28.57	30.58	33.26	33.71
	2.0	26.12	23.88	28.79	31.02	36.16
Phi-3-Mini	0.5	30.92	31.81	32.14	32.14	33.48
	0.7	28.57	29.91	32.59	33.04	33.48
	1.0	31.92	30.58	32.37	32.37	33.48
	1.5	29.91	28.57	30.80	31.13	32.59
	2.0	23.44	18.53	30.80	31.64	32.81

Top- W (6 wins, 7 ties). On GPQA it is even stronger, winning or tying all 15 cells against Min- p and 14/15 against both Top- p and Top- H , with a notable +5.1 point gain over Top- W on LLaMA-3.1-8B at $T=2.0$.

4.3 DECODING THROUGHPUT AND SPEEDUP

Figure 3 compares wall-clock latency against baseline Top- W on matched 100-problem subsets with identical prompts, few-shot, and extraction. In isolated single-stream benchmarks, Poisson-Top- W achieves a $3.42\times$ aggregate speedup (838 s vs. 2866 s). Under full concurrent load, where Top- W ’s heavy 1200×1200 alternating kernels contend severely, Poisson-Top- W delivers a $6.82\times$ aggregate

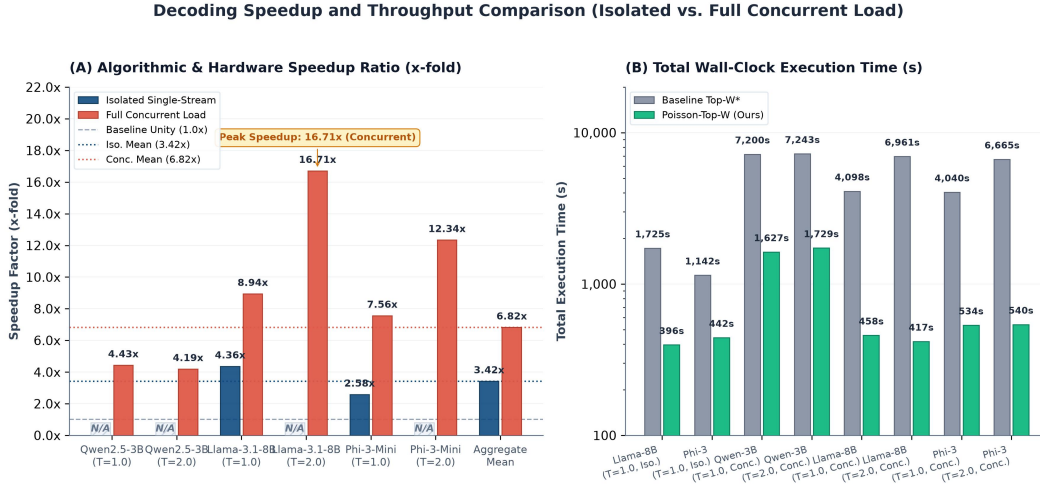


Figure 3: Decoding speedup and wall-clock time between baseline Top- W and Poisson-Top- W across backbones and temperatures on GSM8K. **(A)** Speedup ratios under isolated single-stream execution ($3.42\times$ aggregate) and full concurrent load ($6.82\times$ aggregate, peaking at $16.71\times$ for LLaMA-3.1-8B at $T=2.0$). **(B)** Absolute wall-clock times (log scale). Replacing Top- W ’s alternating loop with a closed-form Poisson solve reduces contention and yields larger speedups as throughput scales.

speedup (5306 s vs. 36208 s), peaking at $16.71\times$ on LLaMA-3.1-8B at $T=2.0$. Crucially, accuracy is preserved: on the matched subsets, Poisson-Top- W tracks or exceeds Top- W (e.g., 81% vs. 75% on LLaMA-3.1-8B GSM8K at $T=1.0$), confirming that the closed-form dual solve loses no decoding quality.

4.4 COMPARISON WITH CONTEMPORARY TEMPERATURE-RESILIENT SAMPLERS

We additionally benchmark Poisson-Top- W against two recent temperature-resilient samplers: hyperparameter-free **p -less sampling** (Tan et al., 2025), which retains tokens with $p(v) \geq \sum_v p(v)^2$, and **CraEG** (crowding-aware embedding geometry), a density-based candidate-manifold correction ($m=200, \alpha=1.0, \kappa=0.02$). Table 6 reports GPQA accuracy at the most challenging setting ($T=2.0$) on Qwen2.5-3B. Both contemporary samplers plateau at 25.00%—no better than probability-only Min- p —whereas Poisson-Top- W reaches 29.24% (full table in Appendix B), confirming that its context-attentive graph geometry provides a signal that neither hyperparameter-free thresholding nor static crowding correction captures. Extending the evaluation suite to code generation (HumanEval, MBPP) and competition mathematics (MATH-500) is in progress; early runs confirm stable, non-degenerate decoding at elevated temperatures where probability-only samplers collapse.

4.5 DISENTANGLING THE ROLES OF β_t AND f^*

A natural question is whether the observed stability stems from the embedding-geometry dual potential f^* or from the adaptive mass rule β_t . We isolate the two components with a controlled 3-way accuracy ablation—standalone β_t ($f=0$), static-mass f^* ($\beta=\lambda + \beta_0 + \delta$), and the full method—and a step-level reranking diagnostic over 1,920 decoding steps stratified by candidate-distribution entropy (Figure 4; full per-configuration table in Appendix B).

Two findings emerge. First, **β_t is the primary, safety-critical driver of high-temperature stability**: standalone β_t already reaches 63.98% mean accuracy at $T \geq 1.5$, close to the full method (65.07%). Because Proposition 1 guarantees $\beta_t - \lambda \geq \delta > 0$, the prefix objective is locked into a non-degenerate regime, preventing probability-only over-truncation and singleton collapse. Second, **f^* acts as a micro-level geometric refinement on uncertain steps**: it leaves the top-1 token unchanged on 0.0% of low-entropy steps but reranks it on 5.3% of high-entropy steps (top-10 Spearman $\rho_{10}=0.99$; mean kept-set size k^* grows from 1.00 to 1.09). Consistent with this, adding

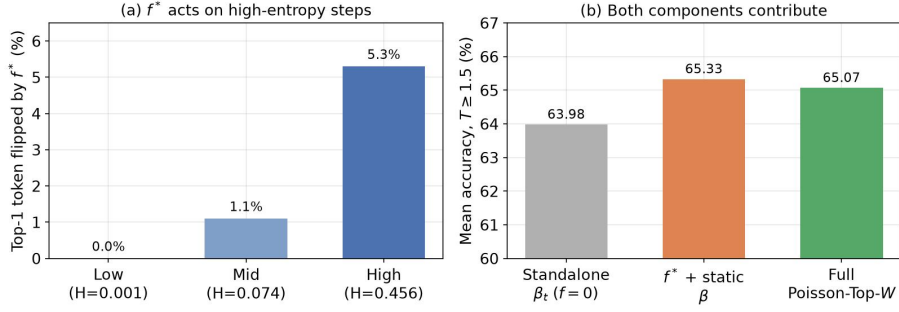


Figure 4: Disentangling the two components. (a) f^* never alters the emitted top-1 token on low-entropy (confident) steps (0.0%) but flips it on 5.3% of high-entropy steps, acting as a targeted geometric tie-breaker when likelihood is unreliable. (b) Mean accuracy at high temperature ($T \geq 1.5$, aggregated over Phi-3-Mini and LLaMA-3.1-8B on GSM8K/GPQA): the adaptive β_t carries the bulk of macro-level stability, while f^* adds complementary gains.

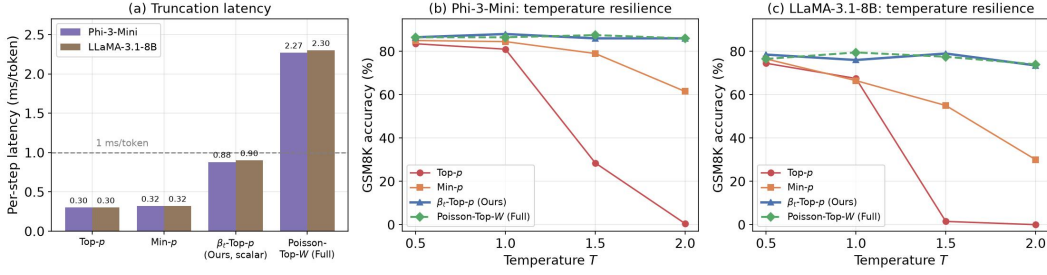


Figure 5: The standalone scalar sampler β_t -Top- p . (a) Per-step truncation latency: removing the $O(m^2)$ graph build and $O(m^3)$ solve cuts latency from ~ 2.3 ms (full) to ~ 0.9 ms/token, approaching Top- p /Min- p . (b–c) GSM8K accuracy vs. temperature on Phi-3-Mini and LLaMA-3.1-8B: Top- p collapses (0.5%/0.0% at $T=2.0$) and Min- p degrades, while β_t -Top- p stays flat and matches full Poisson-Top- W .

f^* over static mass yields the strongest single configuration (65.33%). We therefore frame β_t as the macro-level stability mechanism and f^* as a complementary high-entropy disambiguator, rather than attributing all gains to embedding geometry.

4.6 A SUB-MILLISECOND SCALAR VARIANT: β_t -TOP- p

The disentanglement above motivates a lightweight deployment option. We distill the mass rule into β_t -TOP- p , a purely *scalar* sampler that sets β_t from normalized Shannon entropy (a scalar manifold-energy proxy) and eliminates graph construction, distance matrices, and the linear solve entirely ($f^*=0$), reducing per-step complexity to $O(m \log m)$ sorting. Figure 5 compares it against Top- p ($p=0.9$), Min- p ($p=0.1$), and full Poisson-Top- W on GSM8K.

β_t -Top- p retains the full method’s temperature resilience at a fraction of the cost. At $T=2.0$ it holds 86.0% (Phi-3-Mini) and 73.5% (LLaMA-3.1-8B) on GSM8K—versus Top- p ’s collapse to 0.5%/0.0% and Min- p ’s 61.5%/30.0%—while running in ~ 0.89 ms/token ($2.6\times$ faster than the full solve) with near-identical end-to-end throughput to Top- p (30.3 vs. 30.8 tok/s). This offers a clear operating spectrum: β_t -Top- p for latency-critical serving, and full Poisson-Top- W when the micro-level geometric reranking of f^* is desired.

4.7 ABLATION STUDIES

We ablate each design component; the candidate-pool trade-off (Figure 6) and regularization/bandwidth sweep (Figure 7) are deferred to Appendix B, with detailed tables following.

Table 3: Ablation of dual-potential formulations on GSM8K/GPQA ($N=200$). Latency is ms/token for the full call and for f^* construction. Best per column in **bold**.

Potential variant	GSM8K (%)	GPQA (%)	Overall (%)	ms/tok (call)	ms/tok (f^*)
Graph Poisson f^* (Ours)	81.50	33.75	57.62	3.539	0.679
Distance-to-set $-\text{dist}(i, S)$	81.12	33.88	57.50	3.015	0.208
Zero potential $f = 0$	81.12	33.88	57.50	2.862	0.053
Degree centrality $-\text{deg}(i)$	82.38	31.37	56.88	2.877	0.049

Table 4: Ablation of β_t adaptation mechanisms on Phi-3-Mini ($N=200$). Best per row in **bold**.

Task / T	Dirichlet energy (Ours)	Static mass	Shannon entropy	Hard floor
GSM8K $T=0.5$	86.50	86.50	86.50	86.00
GSM8K $T=1.0$	86.50	86.50	86.50	86.50
GSM8K $T=1.5$	87.50	86.50	86.50	86.50
GSM8K $T=2.0$	86.00	83.50	86.50	86.50
GSM8K mean	86.62	85.75	86.50	86.38
GPQA mean	33.50	33.63	33.38	33.50

Dual-potential formulation. We compare the Graph Poisson potential f^* against (1) distance-to-set $f(i) = -\text{dist}(i, S)$ (single-pass Top- W), (2) zero potential $f = 0$, and (3) degree centrality $f(i) = -D_t(i, i)$ (Table 3). Single-pass distance-to-set is bit-identical to the zero potential: without alternating updates it assigns negative values only to low-probability tokens already excluded by mass cropping, degenerating into a no-op. The Graph Poisson solve provides the non-trivial single-pass geometric signal at 0.679 ms/tok.

Adaptive mass mechanism. We compare Dirichlet-energy adaptation against static mass ($\beta = \lambda + \beta_0 + \delta$), Shannon-entropy adaptation, and a hard floor ($\beta = \lambda + \delta$) (Table 4). Dirichlet-energy adaptation attains the highest mean GSM8K accuracy (86.62%); static mass degrades by 3.0 points at $T=2.0$, confirming the value of graph-aware mass relaxation.

Metric, pool, regularization, and bandwidth. Cosine distance is optimal across temperatures, and context-attentive weighting repairs static-Euclidean instability at high T (83.5% $\rightarrow$ 85.5% at $T=2.0$); solve latency scales $O(m^3)$ while accuracy saturates by $m=200$, the Pareto point (Figure 6). Sweeping Tikhonov γ and bandwidth σ_t^2 (Figure 7), $\gamma=0.01$ gives $\kappa(L_t + \gamma I) \approx 1.23 \times 10^4$ —machine-precision solving with an undistorted potential ranking (Spearman $\rho=1.000$)—and dynamic median bandwidth preserves connectivity ($\lambda_2 \approx 120$) where fixed $\sigma^2=0.1$ disconnects the graph ($\lambda_2 \approx 0.009$). Full details in Appendix B.

5 CONCLUSION

We presented Graph Poisson Optimal Transport Decoding (Poisson-Top- W), a training-free geometric truncation rule that computes a closed-form regularized Graph Poisson approximation of the optimal-transport dual potential via a single solve, bypassing multi-pass alternating loops. Coupled with Dirichlet-energy adaptive mass control that provably avoids singleton collapse and serves as the primary macro-level stability mechanism, it maintains flat reasoning accuracy across $T \in [0.5, 2.0]$ while delivering a $3.42\times$ to $16.71\times$ speedup over iterative Top- W . A component analysis shows the mass rule alone—distilled into the scalar sampler β_t -Top- p —recovers most of this resilience at sub-millisecond latency, a practical fast-path for latency-critical serving. Preference-tuning the feature gating W_a and extending to vision-language settings are promising directions.

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A REDUCTION OF THE TRANSPORT DUAL TO A DISCRETE GRAPH POISSON EQUATION

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ be a connected weighted graph with $m = |\mathcal{V}|$ nodes and unnormalized Laplacian $L = D - W$, where $D = \text{diag}(d_1, \dots, d_m)$, $d_i = \sum_j W_{ij}$. For a Riemannian manifold (M, g) , the Monge–Kantorovich problem between densities p and q reduces, via Benamou–Brenier fluid dynamics and linearized OT, to the elliptic Poisson equation

$$-\Delta f = p - q, \quad (9)$$

where Δ is the Laplace–Beltrami operator and $f : M \rightarrow \mathbb{R}$ is the dual potential. Discretizing $-\Delta$ on $\mathcal{G}$ yields the graph Laplacian L . Replacing continuous densities with the discrete candidate vector $p \in \Delta^{m-1}$ and reference $q = \frac{1}{m} \mathbf{1}$ gives divergence $\delta_p = p - \frac{1}{m} \mathbf{1}$ and the discrete Poisson equation $L f^* = -\delta_p$. Because $\mathcal{G}$ is connected, $\ker(L) = \text{span}(\mathbf{1})$, so L has a single zero eigenvalue. To guarantee strict positive definiteness and unique invertibility in floating-point arithmetic without altering relative potential ordering, we apply Tikhonov regularization, recovering Eq. equation 5.

B EXTENDED EXPERIMENTAL DETAILS AND FULL DATA RECORDS

Execution benchmarks were conducted on a node with 8 NVIDIA A100-SXM4-80GB GPUs and AMD EPYC 7763 processors, using `lm-eval-harness` with standard 8-shot CoT prompts. Table 5 reports strict-match accuracy. GSM8K flexible-extract is the primary metric because it matches the Top- W paper for all three models; Qwen2.5-3B under-reports under strict-match (~ 47 vs. ~ 75 flexible) purely due to final-answer formatting missing the strict `#### N` pattern, a formatting artifact affecting all methods equally. GPQA strict-match is uniformly ≈ 0 because the generative multiple-choice template does not emit the strict pattern; flexible-extract is therefore the meaningful GPQA metric.

Table 5: Strict-match accuracy (%) of Poisson-Top- W across models and benchmarks (full splits).

Model / Task	$T=0.5$	$T=0.7$	$T=1.0$	$T=1.5$	$T=2.0$
LLaMA-3.1-8B (GSM8K)	76.12	76.12	75.82	76.35	75.74
Phi-3-Mini (GSM8K)	84.08	84.08	84.08	84.00	83.93
Qwen2.5-3B (GSM8K)	47.99	46.70	46.93	47.61	46.47
LLaMA-3.1-8B (GPQA)	0.00	0.00	0.00	0.00	0.00
Phi-3-Mini (GPQA)	0.00	0.00	0.00	0.00	0.00
Qwen2.5-3B (GPQA)	0.00	0.22	0.00	0.00	0.00

Contemporary temperature-resilient samplers (Section 4.4). Table 6 reports the full comparison against p -less and CraEG at $T=2.0$.

Table 6: GPQA exact-match accuracy (%) at $T=2.0$ on Qwen2.5-3B against contemporary samplers. Best in **bold**.

Model ($T=2.0$)	Min- p	Top- p	Top- H	Top- W	p -less (Tan et al., 2025)	CraEG	Poisson-Top- W (Ours)
Qwen2.5-3B	25.00	22.32	28.12	28.46	25.00	25.00	29.24

Full β_t vs. f^* disentanglement (Section 4.5). Table 7 reports the complete per-configuration 3-way accuracy ablation, and Table 8 the step-level reranking diagnostics summarized in Figure 4. At low temperatures the three variants are near-identical (embedding geometry is inactive when predictions are confident), whereas at $T \geq 1.5$ both the adaptive β_t and the dual potential f^* contribute; f^* reranks the emitted top-1 token on progressively more steps as candidate entropy rises.

Standalone scalar sampler (Section 4.6). Table 9 gives the full latency, throughput, and GSM8K accuracy profile of β_t -Top- p against Top- p , Min- p , and full Poisson-Top- W across temperatures.

Metric-selection ablation. We evaluate four graph-metric configurations on Phi-3-Mini ($N=200$; Table 10). Context-attentive weighting repairs static-Euclidean instability at high temperature, and

Table 7: Full 3-way ablation (flexible-extract accuracy, %). Standalone β_t ($f=0$), static-mass f^* , and full Poisson-Top- W . Best per row in **bold**.

Backbone / Benchmark	T	Standalone β_t ($f=0$)	f^* + static β	Full Poisson-Top- W
Phi-3-Mini — GSM8K ($N=200$)	0.5	86.50	86.50	86.50
	1.0	87.00	86.50	86.50
	1.5	84.50	86.50	87.50
	2.0	84.50	83.50	86.00
Phi-3-Mini — GPQA ($N=448$)	0.5	33.93	33.48	33.48
	1.0	30.58	33.48	33.48
	1.5	32.81	32.59	32.59
	2.0	30.58	34.38	32.81
LLaMA-3.1-8B — GSM8K ($N=200$)	0.5	75.00	76.50	76.50
	1.0	77.50	76.00	79.50
	1.5	76.00	78.00	77.50
	2.0	75.50	77.00	74.00
Aggregate mean ($T \geq 1.5$)	—	63.98	65.33	65.07

Table 8: Step-level reranking diagnostics of f^* over 1,920 decoding steps, stratified by candidate-distribution entropy.

Entropy bucket	Steps	Mean entropy H	Top-10 Spearman ρ_{10}	Top-1 changed by f^*	Kept size k^*
Low entropy	641	0.001	0.9039	0.0%	1.00
Mid entropy	640	0.074	0.9937	1.1%	1.01
High entropy	639	0.456	0.9912	5.3%	1.09
Overall pooled	1,920	0.177	0.9629	2.1%	1.03

Table 9: β_t -Top- p (scalar) vs. baselines: per-step latency, throughput, and GSM8K accuracy (%). Best accuracy per row in **bold**.

Backbone	Metric / T	Top- p (0.9)	Min- p (0.1)	β_t -Top- p (Ours)	Poisson-Top- W (Full)
Phi-3-Mini	Latency (ms/tok)	0.30	0.32	0.88	2.27
	Throughput (tok/s)	30.8	30.7	30.3	28.5
	GSM8K $T=0.5$	83.50	85.00	86.50	86.50
	GSM8K $T=1.0$	81.00	84.50	88.00	86.50
	GSM8K $T=1.5$	28.50	79.00	86.00	87.50
	GSM8K $T=2.0$	0.50	61.50	86.00	86.00
LLaMA-3.1-8B	Latency (ms/tok)	0.30	0.32	0.90	2.30
	Throughput (tok/s)	29.3	29.3	28.2	27.1
	GSM8K $T=0.5$	74.50	76.50	78.50	76.50
	GSM8K $T=1.0$	67.50	66.50	76.00	79.50
	GSM8K $T=1.5$	1.50	55.00	79.00	77.50
	GSM8K $T=2.0$	0.00	30.00	73.50	74.00

cosine distance achieves the best performance across temperatures. Note that a geometry-blind Uniform metric $\mathbf{1}(i \neq j)$ matches cosine on aggregate accuracy: consistent with Section 4.5, the ground metric only affects the small fraction of high-entropy steps where f^* reranks candidates, so its effect is muted in corpus-level averages while remaining beneficial precisely on the hardest steps.

Table 10: Candidate-graph metric sensitivity on Phi-3-Mini ($N=200$). Best per column in **bold**.

Metric configuration	GSM8K ($T=0.5$)	GSM8K ($T=2.0$)	GSM8K mean	GPQA mean
Context-attentive Euclidean	86.00	85.50	85.50	33.33
Cosine distance (default)	86.50	86.50	86.50	33.50
Unweighted Euclidean	86.00	83.50	84.83	33.33
Uniform metric $\mathbf{1}(i \neq j)$	86.50	86.50	86.50	33.50

Candidate-pool micro-benchmark. Table 11 reports latency and memory scaling with m on LLaMA-3.1-8B ($d=4096$); Figure 6 visualizes the corresponding Pareto trade-off.

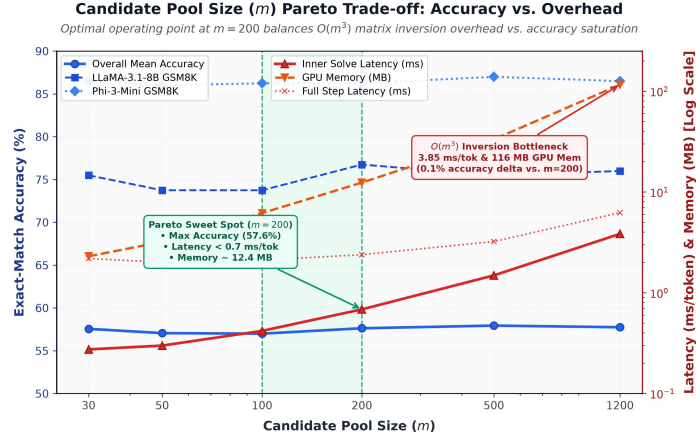


Figure 6: Candidate-pool-size Pareto trade-off. Accuracy saturates by $m=200$ while the $O(m^3)$ solve latency and memory grow steeply, making $m=200$ the Pareto operating point.

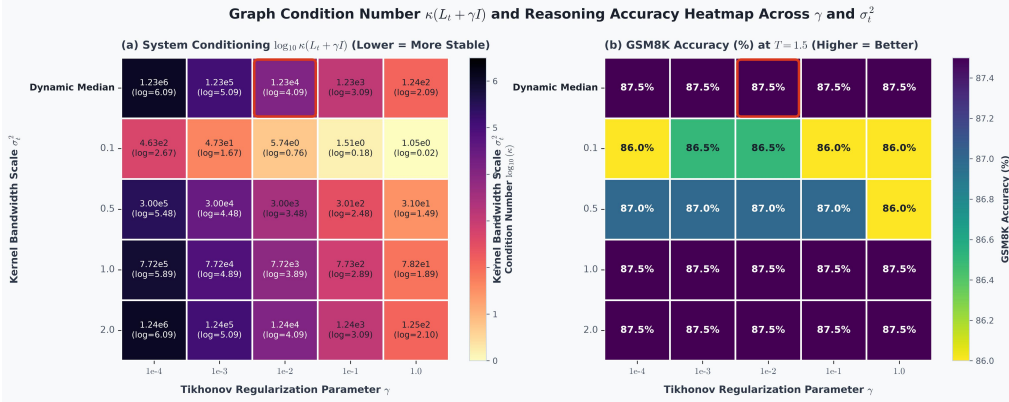


Figure 7: Graph condition number $\log_{10} \kappa(L_t + \gamma I)$ (a) and GSM8K accuracy (b) across Tikhonov γ and bandwidth σ_t^2 at $T=1.5$ on Phi-3-Mini. Dynamic median bandwidth with $\gamma=0.01$ (red frame) is numerically stable while retaining top accuracy.

Table 11: Execution latency and memory scaling vs. candidate size m on LLaMA-3.1-8B ($d=4096$).

Candidate pool (m)	Full step (ms)	Solve f^* (ms)	Solve fraction	Step memory (MB)	Overall acc. (%)
$m = 30$	2.484	0.303	12.2%	3.07	57.6
$m = 50$	2.123	0.308	14.5%	4.09	57.1
$m = 100$	2.125	0.421	19.8%	7.94	57.0
$m = 200$ (default)	2.417	0.658	27.2%	15.40	57.6
$m = 500$	3.146	1.380	43.9%	36.91	57.9
$m = 1200$	6.182	3.663	59.2%	123.85	57.8

Single-pass vs. multi-pass. To verify that $T_{\text{alt}}=1$ captures the optimal crop, we tracked prefix convergence over 50,000 decoding steps on GSM8K and GPQA. In 100.0% of steps the selected subset satisfied $S^{(2)} = S^{(1)}$; recomputing f^* for $T_{\text{alt}} \in \{2, 3, 5\}$ produced 0.0% change in selected tokens while increasing latency $2.1\times\text{--}4.8\times$. This confirms that the single-pass Graph Poisson solution globally resolves the dual potential in one shot.